الأسئلة المقالية — 11 متقدم · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (36 سؤالاً)

LO21 عبارات الأساس الطبيعي واللوغاريتم الطبيعي · Evaluate expressions involving the natural base and

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21.1 LO21 · 10

(a) Graph the function. (b) State the domain and range. (c) Find its average rate of change on [-5, -2].

$$f(x) = 3e^{x-1} + 3$$

a. Graph: key features to read from your sketch

b. Domain and range

c. Average rate of change

21.2 LO21 · 11

(a) Graph the function. (b) State the domain and range. (c) Find its average rate of change on [-3, -1].

$$f(x) = 4e^{2x} - 1$$

a. Graph: key features to read from your sketch

b. Domain and range

c. Average rate of change

21.3 LO21 · 12

(a) Graph the function. (b) State the domain and range. (c) Find its average rate of change on [-7, -4].

$$f(x) = -2e^{x+3} + 2$$

a. Graph: key features to read from your sketch

b. Domain and range

c. Average rate of change

21.4 LO21 ·

For the given function: (a) graph it; (b) state the domain and range; (c) find the average rate of change over [0,2].

$$f(x) = 2e^x - 3$$

a. Graph: key features to read from your sketch

b. Domain and range

c. Average rate of change

21.5 LO21 ·

For the given function: (a) graph it; (b) state the domain and range; (c) find the average rate of change over [0,2].

$$f(x) = 3e^x - 4$$

a. Graph: key features to read from your sketch

b. Domain and range

c. Average rate of change

21.7 LO21 ·

Evaluate the expression exactly.

$$\ln\left(e^5\right) + e^{\ln 3}$$

a. Show all steps and check your answer.

21.7 LO21 ·

Solve the equation. Round to four decimal places.

$$e^x = 7$$

a. Show all steps and check your answer.

22.1 LO22 · thorium

Thorium-230 has a half-life of 75,381 years. (a) Determine the positive decay constant k in y=a e^(−kt) and write the decay equation. (b) How much remains of a 2 g sample after 1,500 years?

a. Decay constant and model

b. Remaining mass

22.2 LO22 · iodine

Iodine-131 has a half-life of 8.02 days. (a) Write its continuous exponential decay equation; round k to the nearest thousandth. (b) How much of a 15 g sample remains after 20 days? Use the unrounded constant for the final mass.

a. Decay equation

b. Remaining mass

22.3 LO22 ·

A substance has half-life 6 days and initial mass 100 g. (a) Write a continuous decay model $M(t)=M_0e^{(-kt)}$. (b) Find the mass after 18 days. (c) How long until only one quarter remains?

a. Decay model

b. Remaining mass

c. Time for a fraction

22.4 LO22 ·

A substance has half-life 8 days and initial mass 120 g. (a) Write a continuous decay model $M(t)=M_0e^{(-kt)}$. (b) Find the mass after 24 days. (c) How long until only one quarter remains?

a. Decay model

b. Remaining mass

c. Time for a fraction

22.5 LO22 ·

A substance has half-life 10 days and initial mass 140 g. (a) Write a continuous decay model $M(t)=M_0e^{(-kt)}$. (b) Find the mass after 30 days. (c) How long until only one quarter remains?

a. Decay model

b. Remaining mass

c. Time for a fraction

22.7 LO22 ·

A population of 500 grows continuously at 4% per year, P(t) = 500e^{0.04t}. How many years until it doubles? Round to the nearest tenth.

a. Show all steps and check your answer.

22.7 LO22 ·

A culture grows from 200 to 800 bacteria in 6 hours, N(t) = 200e^{kt}. Find the growth rate k. Round to four decimal places.

a. Show all steps and check your answer.

23.1 LO23 · example

Find all zeros and asymptotes, then graph the function. Indicate any excluded points.

$$f(x) = \frac{x^2+2x+1}{x+4}$$

a. Zeros

b. Asymptotes and excluded points

c. Graph (sketch on paper using the zeros, asymptotes and intercept)

23.2 LO23 · 11

Find all zeros and asymptotes, then graph the function. Indicate any excluded points.

$f(x) = \frac{(x-4)^2}{x+2}$

a. Zeros

b. Asymptotes and excluded points

c. Graph (sketch on paper using the zeros, asymptotes and intercept)

23.3 LO23 · 12

Find all zeros and asymptotes, then graph the function. Indicate any excluded points.

$f(x) = \frac{(x+3)^2}{x-5}$

a. Zeros

b. Asymptotes and excluded points

c. Graph (sketch on paper using the zeros, asymptotes and intercept)

23.4 LO23 · 13

Find all zeros and asymptotes, then graph the function. Indicate any excluded points.

$f(x) = \frac{6x^2+4x+2}{x+2}$

a. Zeros

b. Asymptotes and excluded points

c. Graph (sketch on paper using the zeros, asymptotes and intercept)

23.5 LO23 · 14

Find all zeros and asymptotes, then graph the function. Indicate any excluded points.

$f(x) = \frac{2x^2+7x}{x-2}$

a. Zeros

b. Asymptotes and excluded points

c. Graph (sketch on paper using the zeros, asymptotes and intercept)

23.6 LO23 · 15

Find all zeros and asymptotes, then graph the function. Indicate any excluded points.

$f(x) = \frac{3x^2+8}{2x-1}$

a. Zeros

b. Asymptotes and excluded points

c. Graph (sketch on paper using the zeros, asymptotes and intercept)

23.7 LO23 · 16

Find all zeros and asymptotes, then graph the function. Indicate any excluded points.

$f(x) = \frac{2x^2+5}{3x+4}$

a. Zeros

b. Asymptotes and excluded points

c. Graph (sketch on paper using the zeros, asymptotes and intercept)

23.8 LO23 · hole

Find the domain, zeros, vertical and horizontal asymptotes, and any hole; then sketch the graph.

$f(x) = \frac{x^2-2x-24}{2x^3+6x^2-8x}$

a. Domain and hole

b. Zeros and asymptotes

c. Graph

24.1 LO24 · empirical

A normal distribution has mean 42 and standard deviation 6. Using the Empirical Rule: (a) Find the interval containing the middle 95% of the data. (b) What percentage is greater than 36?

a. Middle 95%

b. Percentage above 36

24.2 LO24 · traffic

Daily visits to a news website are normally distributed with mean 98,452 and standard deviation 10,325. Find the probability of at least 100,000 visits. Use the supplied standard normal table.

a. Standardize

b. Find tail probability

24.3 LO24 · 14

Test scores are normally distributed with mean 100 and standard deviation 12.3. (a) What percentage lies between 70 and 80? (b) What percentage is over 115? (c) Out of 75 people, how many are expected to score below 75? Use the supplied standard normal table.

a. Between 70 and 80

b. Over 115

c. Expected number below 75

24.4 LO24 · book-service

Customer-service waiting time is normally distributed with mean 21.3 s and standard deviation 3.8 s. What percentage of waits exceed 30 s? Use the supplied standard normal table.

a. Standardize

b. Upper-tail percentage

24.5 LO24 ·

Scores follow a normal distribution with mean 70 and standard deviation 8. (a) Find the score at the 90th percentile, to one decimal place. (b) Find P(62<X<78). Use the supplied standard normal table.

a. Inverse normal value

b. Interval probability

25.1 LO25 · example

A roundabout has diameter 160 ft. Find the distance a car travels through three-fourths of a full turn. State the central angle in degrees and radians.

a. Angle conversion

b. Arc length

25.2 LO25 · 52

A roundabout has diameter 200 m. Find the distance travelled through one-fourth of a full turn. State the central angle in degrees and radians.

a. Angle conversion

b. Arc length

25.3 LO25 · 53

The minute hand of a clock is 5 in long. How far does its tip travel from 12:00 to 12:40? State the rotation in degrees and radians.

a. Angle conversion

b. Arc length

25.4 LO25 · trig7

State amplitude, period and phase shift. Graph one full period and state the domain and range.

$$y = 3\cos(\theta - 45^\circ)$$

a. Amplitude, period and phase shift

b. Graph one full period

c. Domain and range

25.5 LO25 · trig8

State amplitude, period and phase shift. Graph one full period and state the domain and range.

$$y = 2\sin(\theta + 60^\circ)$$

a. Amplitude, period and phase shift

b. Graph one full period

c. Domain and range

25.6 LO25 · trig9

State amplitude, period and phase shift. Graph one full period and state the domain and range.

$$y = \frac{1}{2}\sin(3(\theta - \pi/3))$$

a. Amplitude, period and phase shift

b. Graph one full period

c. Domain and range

25.7 LO25 · trig10

State amplitude, period and phase shift. Graph one full period and state the domain and range.

$$y = \frac{1}{2}\cos(3(\theta - \pi/2))$$

a. Amplitude, period and phase shift

b. Graph one full period

c. Domain and range

25.8 LO25 · trig11

State amplitude, period and phase shift. Graph one full period and state the domain and range.

$$y = 2\cos(\theta + 90^\circ)$$

a. Amplitude, period and phase shift

b. Graph one full period

c. Domain and range

25.9 LO25 · trig12

State amplitude, period and phase shift. Graph one full period and state the domain and range.

$$y = \frac{1}{4}\sin(\theta - 30^\circ)$$

a. Amplitude, period and phase shift

b. Graph one full period

c. Domain and range